

PayFlow on DOKS

Deployment Setup Guide | Saravanan Ramesh | April 2026

A containerized Node.js invoicing application deployed on **DigitalOcean Kubernetes (DOKS)** with PostgreSQL, load balancing, horizontal pod autoscaling, pod anti-affinity for high availability, and admin-protected write operations via Kubernetes Secrets.

Stack: Node.js 18 + Express · PostgreSQL 15 · Docker · Kubernetes · NGINX Ingress · DO Load Balancer · HPA · Pod AntiAffinity

Prerequisites

DigitalOcean account	cloud.digitalocean.com — with credits or billing configured
doctl CLI	Installed and authenticated via doctl auth init
kubectl	Configured after cluster creation
Docker	For building and pushing container images
Helm	For NGINX Ingress and Metrics Server installation
Node.js 18+	Required for local development only

Project Structure

GitHub: github.com/SaravananRamesh/PayFlow

```
payflow-saas/
├── app/
│   ├── server.js # Express app with invoice API and /metrics endpoint
│   ├── Dockerfile # Non-root Alpine image
│   └── package.json
├── k8s/
│   ├── deployment.yaml # PayFlow app deployment with pod anti-affinity
│   ├── service.yaml # ClusterIP service
│   ├── ingress.yaml # NGINX ingress (ingressClassName: nginx)
│   ├── hpa.yaml # HPA – min 2 / max 10 pods / CPU 70%
│   ├── postgres.yaml # PostgreSQL deployment + ClusterIP service
│   ├── pvc.yaml # PersistentVolumeClaim (1Gi DO Block Storage)
│   ├── configmap.yaml # Non-sensitive config (PORT, DB_HOST, DB_NAME, DB_USER)
│   └── secret.yaml # Secret template (ADMIN_PASSWORD, DB_PASSWORD)
└── README.md
```

Step-by-Step Deployment Guide

1 Create a DOKS Cluster

```
doctl kubernetes cluster create payflow-cluster \  
--region nyc1 \  
--node-pool  
"name=worker-pool;size=s-1vcpu-2gb;count=2;auto-scale=true;min-nodes=2;max-nodes=3" \  
--version latest
```

Wait ~4 minutes for provisioning, then configure kubectl:

```
doctl kubernetes cluster kubeconfig save payflow-cluster  
kubectl get nodes # Verify 2 nodes are Ready
```

2 Set Up DO Container Registry

```
doctl registry create payflow-registry --region nyc3  
doctl registry login  
  
# Creates a dockerconfigjson secret – grants cluster pull access  
# Patches the default service account in all namespaces automatically  
doctl registry kubernetes-manifest | kubectl apply -f -  
  
# Links registry in DO control plane (auto-syncs on token rotation)  
doctl kubernetes cluster registry add payflow-cluster
```

3 Build & Push the Docker Image

Apple Silicon Mac users: DOKS nodes run linux/amd64 — always specify the platform flag.

```
cd app/  
npm install  
docker build --platform linux/amd64 \  
-t registry.digitalocean.com/payflow-registry/payflow:latest .  
docker push registry.digitalocean.com/payflow-registry/payflow:latest
```

4 Install NGINX Ingress Controller

Provisions a DO Load Balancer and routes external HTTP traffic into the cluster

```
helm repo add ingress-nginx https://kubernetes.github.io/ingress-nginx
helm repo update
helm install nginx-ingress ingress-nginx/ingress-nginx \
--set controller.publishService.enabled=true
```

Wait ~60 seconds for the DO Load Balancer to provision an external IP:

```
kubectl get svc nginx-ingress-ingress-nginx-controller -w
# Wait until EXTERNAL-IP is assigned (not pending)
```

5 Install Metrics Server (Required for HPA)

Not pre-installed on DOKS — required for HPA CPU metrics and kubectl top

```
helm repo add metrics-server https://kubernetes-sigs.github.io/metrics-server/
helm upgrade --install metrics-server metrics-server/metrics-server \
--namespace metrics-server --create-namespace

# Verify after ~60 seconds
kubectl top nodes
```

*Alternatively: DO Console → your cluster → **Marketplace** tab → search 'Metrics Server' → Install.*

6 Create Kubernetes Secret

Never commit real credentials to Git — create the secret directly via kubectl

```
kubectl create secret generic payflow-secrets \
--from-literal=ADMIN_PASSWORD=yourchosenpassword \
--from-literal=DB_PASSWORD=yourdbpassword \
--dry-run=client -o yaml | kubectl apply -f -
```

7 Deploy PostgreSQL

```
kubectl apply -f k8s/pvc.yaml
kubectl apply -f k8s/postgres.yaml

# Wait until postgres pod is Running before continuing
kubectl get pods -w
```

8 Deploy the Application

```
kubectl apply -f k8s/configmap.yaml
kubectl apply -f k8s/deployment.yaml
kubectl apply -f k8s/service.yaml
kubectl apply -f k8s/ingress.yaml
kubectl apply -f k8s/hpa.yaml
```

Verify everything is running:

```
kubectl get pods -o wide # 2 PayFlow pods + postgres on separate nodes
kubectl get svc # ClusterIP services
kubectl get ingress # Ingress with external IP
kubectl get hpa # HPA targets showing CPU %
```

9 Get Your External IP & Open the App

```
kubectl get svc nginx-ingress-ingress-nginx-controller \
-o jsonpath='{.status.loadBalancer.ingress[0].ip}'
```

Visit <http://<EXTERNAL-IP>> — you should see the PayFlow dashboard.

Using the Dashboard

The dashboard is **read-only by default**. To add, update, or delete invoices:

- Click the **locked** badge on the New Invoice form
- Enter the **ADMIN_PASSWORD** you configured in Step 6
- The form unlocks — you can now add invoices, mark them as paid, or delete them
- Click **admin mode** to lock back

API Endpoints

Endpoint	Method	Auth	Description
/	GET	Public	Invoice dashboard (HTML)
/health	GET	Public	Health check with pod identity and DB status
/invoices	GET	Public	List all invoices (JSON)
/invoices	POST	Admin	Create invoice {client, amount, due}
/invoices/:id	PATCH	Admin	Update invoice status
/invoices/:id	DELETE	Admin	Delete invoice

/metrics	GET	Public	Prometheus-formatted metrics
/generate-report	GET	Public	CPU-intensive endpoint (for HPA demo)

Testing

Verify High Availability & Pod Anti-Affinity

```
# Confirm app pods are on different nodes
kubectl get pods -o wide
# The NODE column shows different node names for each payflow-app pod
```

Verify Load Balancing

```
# Hit health endpoint repeatedly – pod hostname rotates between requests
for i in $(seq 1 10); do
curl -s http://EXTERNAL-IP/health | jq .pod
done
```

Trigger HPA Autoscaling

```
# Install hey: brew install hey (macOS)
hey -z 120s -c 50 http://EXTERNAL-IP/generate-report

# Watch in separate terminals:
kubectl get hpa -w
kubectl get pods -w
kubectl get nodes -w
```

CPU spikes above 70% → HPA scales pods → pod anti-affinity triggers cluster autoscaler to provision a new node. After load drops, pods and nodes scale back down.

Cost Analysis

Resource	Spec	Monthly Cost
DOKS control plane	Managed, NYC1	\$0 (free)
Worker nodes (2×)	s-1vcpu-2gb each — 1 vCPU, 2GB	\$12/node = \$24
DO Load Balancer	Standard, Layer 4	\$12
Container Registry	Starter tier	\$0 (free)
PVC (PostgreSQL)	1Gi DO Block Storage	~\$0.10

Total baseline		~\$36/mo
Node autoscaling peak (3×)	s-1vcpu-2gb × 3	~\$36 nodes only

vs AWS EKS equivalent: ~\$159/mo — DOKS is 4× cheaper for the same architecture.

Cleanup

```
kubectl delete -f k8s/deployment.yaml
kubectl delete -f k8s/service.yaml
kubectl delete -f k8s/ingress.yaml
kubectl delete -f k8s/hpa.yaml
kubectl delete -f k8s/postgres.yaml
kubectl delete -f k8s/pvc.yaml
kubectl delete -f k8s/configmap.yaml
helm uninstall nginx-ingress
doctl kubernetes cluster delete payflow-cluster --force
doctl registry delete payflow-registry --force
```